Problem A
Bit Shuffle

Time limit: 1 second

A **permutation** π of $\{0, 1, \ldots, k-1\}$ **encodes** a k-bit integer a as follows: bit i of a is moved to bit position $\pi(i)$, for every i. Bit 0 is the least significant bit (lsb). Formally, the encoding of a is the integer b whose bit $\pi(i)$ equals bit i of a for all $0 \le i < k$. Treat both a and b as exactly k bits, including leading zero bits.

You are given n pairs of integers $(a_1, b_1), \ldots, (a_n, b_n)$. We want one fixed permutation π under which b_j is the encoding of a_j for every j. If several permutations qualify, the answer is the lexicographically smallest one: the one with the smallest $\pi(0)$, then the smallest $\pi(1)$, and so on.

Task: Output the lexicographically smallest such permutation π, or -1 if no permutation encodes every a_j as b_j.

Example:

Let $k = 3$ and the pairs be $(3, 6)$ and $(4, 1)$. The numbers in binary, listed from bit 0 to bit 2:

Number	Value	Bit 0	Bit 1	Bit 2
a_1	3	1	1	0
b_1	6	0	1	1
a_2	4	0	0	1
b_2	1	1	0	0

It can be shown that the lexicographically smallest permutation π that encodes each a_j to b_j is: $1\ 2\ 0$. That is, $\pi(0) = 1, \pi(1) = 2, \pi(2) = 0$.

1. Bit 0 of $a_1 = 1$ moves to bit $\pi(0) = 1$ of b_1.
   Bit 0 of $a_2 = 0$ moves to bit $\pi(0) = 1$ of b_2.
2. Bit 1 of $a_1 = 1$ moves to bit $\pi(1) = 2$ of b_1.
   Bit 1 of $a_2 = 0$ moves to bit $\pi(1) = 2$ of b_2.
3. Bit 2 of $a_1 = 0$ moves to bit $\pi(2) = 0$ of b_1.
   Bit 2 of $a_2 = 1$ moves to bit $\pi(2) = 0$ of b_2.

The permutation $2\ 1\ 0$ is also valid, but it is lexicographically larger.

Input

First line: two integers n and k ($1 \le n \le 2 \cdot 10^4, 1 \le k \le 60$).

Next n lines: two integers a_j and b_j ($0 \le a_j, b_j < 2^k$) on the j-th line. The same integer or pair may appear more than once.

Output

Print k space-separated integers $\pi(0), \pi(1), \ldots, \pi(k-1)$: the lexicographically smallest valid permutation. If there is none, print -1.

Sample Input 1	Sample Output 1
2 3 3 6 4 1	1 2 0

Sample Input 2	Sample Output 2
1 2 1 3	-1

Sample Input 3	Sample Output 3
3 3 1 2 2 4 4 1	1 2 0

Sample Input 4	Sample Output 4
3 4 6 12 9 10 6 12	-1